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EXPERIMENT 9

PERFORM WEB APPLICATION VULNERABILITY SCANNING

AIM:

To run an automated vulnerability scan against a web application and identify common security weaknesses such as security misconfigurations and missing HTTP security headers.

ALGORITHM:

1. Start the Kali Linux environment and launch OWASP ZAP.
2. Start the DVWA target application and access it at http://127.0.0.1:8081/.
3. Log in to DVWA and set the security level to Low.
4. In ZAP Quick Start, enter the target URL http://127.0.0.1:8081/.
5. Enable the traditional spider and start the automated scan against the local lab target.
6. Wait until ZAP displays “Attack complete” and indicates that results are available in the Alerts tab.
7. Open the Alerts tab and review the findings and their risk levels.
8. Select alerts to inspect the details and affected URLs where available.
9. Generate the ZAP report in HTML format.
10. Record the observed findings and report as the output of the experiment.

TOOLS REQUIRED:

- Kali Linux
- OWASP ZAP (Zed Attack Proxy)
- DVWA running locally in Docker

OUTPUT:

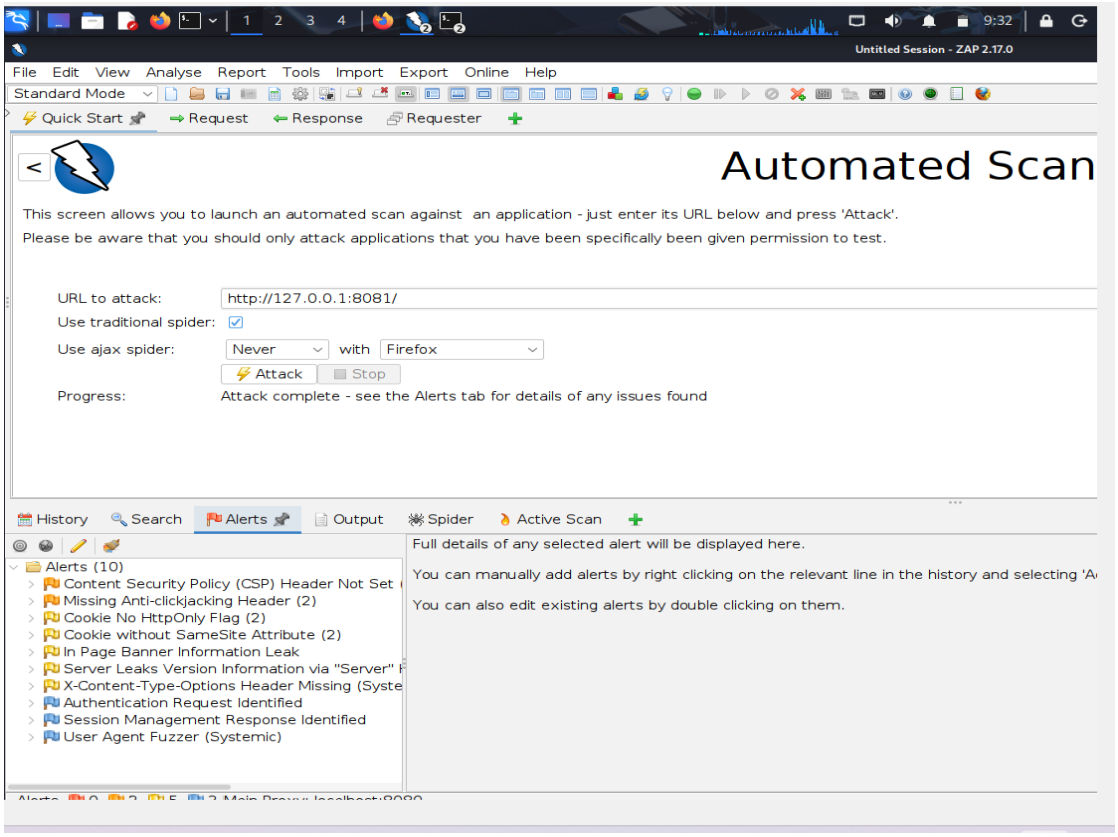


Fig 1: OWASP ZAP automated scan completed for the local DVWA target; the Alerts panel shows 10 alerts.

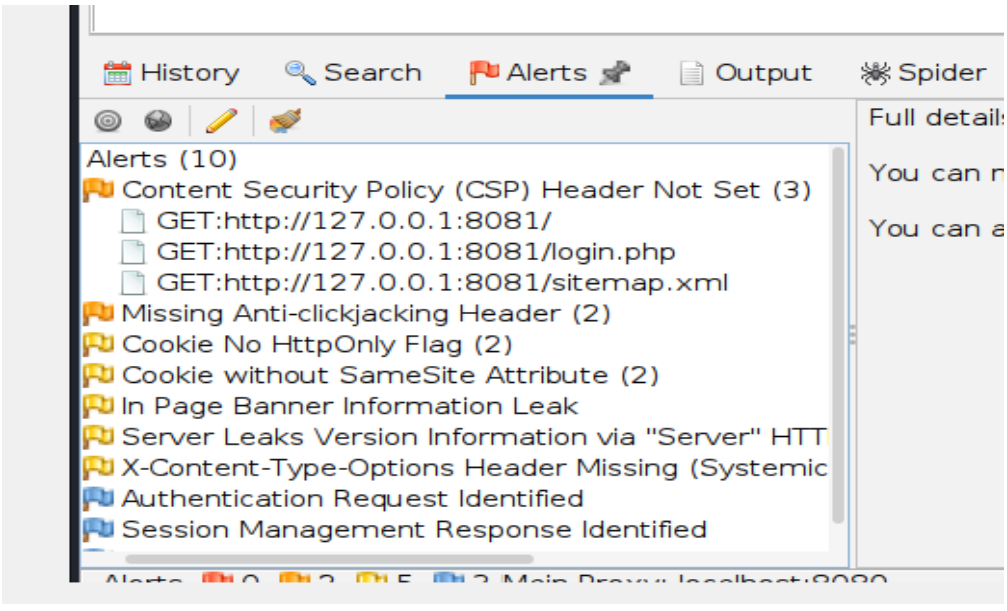


Fig: Actual ZAP Alerts tab showing the scan's detected alert categories, including CSP Header Not Set and missing security headers.

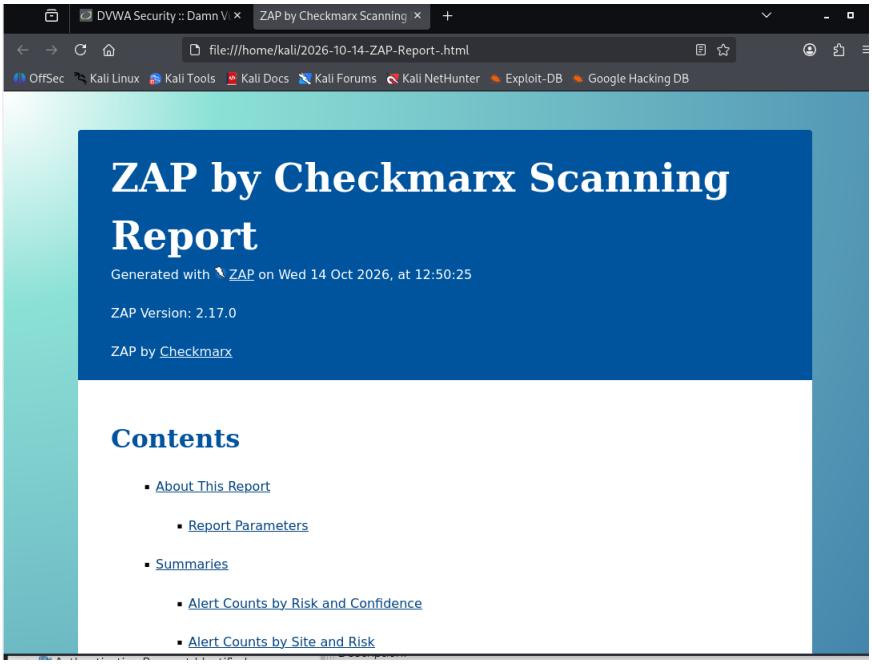


Figure 3: Actual ZAP HTML report opened in the browser, showing the report title, generation time, and ZAP version.

RESULT:

The local DVWA web application was scanned successfully using OWASP ZAP. The completed scan displayed 10 alerts, including a Medium-risk Content Security Policy (CSP) Header Not Set finding, Low-risk missing security-header and cookie-configuration findings, and Informational observations. The ZAP HTML report was generated successfully.